

Program : Diploma in Information Technology	
Course Code : 4269	Course Title: Python Web Application Development Lab
Semester : 4	Credits: 0
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide practical knowledge to develop and deploy Web Applications using Python Frameworks.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basic IT Skills		Introduction to IT systems	1
Knowledge of SQL and its usage		Database Management System lab	3
Python Programming skill		Application Development Lab - I	3
Web technology skills		Fundamentals of Web Development	3

Course Outcomes

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop front-end of a web application.	10	Applying
CO2	Develop a simple web application with form	10	Applying
CO3	Develop a web application with multiple forms	10	Applying
CO4	Make use of front-end framework to develop a web application.	12	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

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Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop front end of a web application		
M1.01	Develop simple web page using HTML5, CSS3 and JavaScript.	4	Applying
M1.02	Develop simple web page using Bootstrap.	3	Applying
M1.03	Develop simple web page using Bootstrap and jQuery.	3	Applying
CO2	Develop a simple web application with form		
M2.01	Practice installation of Python IDE	2	Applying
M2.02	Practice installation of Python Framework	2	Applying
M2.03	Practice installation of database server	2	Applying
M2.04	Develop simple web application with form.	2	Applying
M2.05	Develop simple web application with form using template.	2	Applying
	Lab Exam – I	1.5	
CO3	Develop a web application with multiple forms		

M3.01	Develop a web application with multiple forms (Like Students Attendance Management, Library Management)	10	Applying
CO4	Make use of front-end framework to develop a web application.		
M4.01	Make use of front-end framework to develop a web application (Like Point of Sale, Payroll Management)	3	Applying
M4.02	Practice deployment of web application in a web server or cloud server.	3	Applying
M4.03	Open-ended experiment. **	6	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Inventory Management
2. Hospital Management
3. Out Patient Management

Text / Reference:

T/R	Book Title / Author
T1	Laura Lemay, Rafe Colburn, Jennifer Kyrnin, “ <i>Mastering HTML, CSS & Javascript Web Publishing</i> ”, BPB Publications 2016
T2	Miguel Grinberg, “ <i>Flask Web Development: Developing Web Applications with Python</i> ”, O'Reilly 2018

Online Resources:

Sl. No.	Website Link
1	https://www.tutorialrepublic.com/html-tutorial/
2	https://www.tutorialrepublic.com/css-tutorial/
3	https://www.tutorialrepublic.com/javascript-tutorial/

4	https://www.tutorialrepublic.com/jquery-tutorial/
5	https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/
6	https://www.tutorialrepublic.com/sql-tutorial/
7	https://www.tutorialspoint.com/flask/index.htm
8	https://www.javatpoint.com/flask-tutorial